FAKE NEWS DETECTION USING MACHINE LEARNING AND DATA MINING

Abstract

The rapid expansion of online news platforms and social media has increased the spread of misinformation and fake news, creating significant challenges for information credibility and public trust. Manual verification of news articles is often time-consuming and inefficient, making automated detection systems essential. This study presents a machine learning-based fake news detection framework utilizing Data Mining and Natural Language Processing (NLP) techniques. A multilingual news dataset containing both fake and real news articles was collected from the Mendeley Data Repository. The textual data underwent preprocessing operations including lowercasing, punctuation removal, tokenization, stop-word elimination, and stemming. TF-IDF vectorization was employed for feature extraction, while Logistic Regression was used as the classification algorithm. Experimental results demonstrated an overall classification accuracy of 95.95%, with balanced precision, recall, and F1-scores of approximately 96%. The findings indicate that lightweight machine learning approaches remain highly effective for fake news detection while requiring significantly lower computational resources than many deep learning models.

Keywords: Fake News Detection, Machine Learning, Data Mining, Natural Language Processing, TF-IDF, Logistic Regression.

1. Introduction

The internet has revolutionized the way information is generated, distributed, and consumed. News articles are now accessible instantly through websites, blogs, and social media platforms. However, the ease of information sharing has also contributed to the rapid dissemination of fake news and misinformation

The widespread circulation of fake news can have serious consequences, including political manipulation, economic instability, social polarization, and public distrust in credible information sources. Traditional fact-checking methods rely heavily on human experts and manual verification processes, which are often slow and incapable of handling the vast volume of online content produced daily.

Machine Learning and Data Mining techniques provide an effective solution for automatically identifying deceptive news content. By learning patterns from previously labeled examples, machine learning models can classify unseen news articles as either fake or real with high accuracy.

This project proposes an automated fake news detection framework that combines Natural Language Processing techniques, TF-IDF feature extraction, and Logistic Regression classification. The primary objective is to develop a reliable, scalable, and computationally efficient system capable of distinguishing fake news from authentic news articles in real-world environments.

2. Methodology

2.1 Purpose

The purpose of this project is to develop a machine learning framework capable of automatically detecting fake news articles from textual content. The system aims to provide accurate predictions while remaining computationally efficient and suitable for practical deployment.

2.2 Research Design

The proposed system follows a supervised machine learning approach involving:

- Data collection from a publicly available dataset
- Data preprocessing and cleaning
- Feature extraction using TF-IDF
- Model training using Logistic Regression
- Performance evaluation using classification metrics

2.3 Dataset Description

The dataset was obtained from the Mendeley Data Repository and consists of two files:

- Fake.xlsx – Fake News Articles
- Real.xlsx – Authentic News Articles

Dataset Statistics

Attribute	Value
Fake News Records	4,912
Real News Records	5,294
Total Records	10,206
Features	10 Columns
Classification Type	Binary Classification

Class Labels:

- Real News = 0

- Fake News = 1

The balanced distribution of classes reduces classification bias and improves model reliability.

2.4 Tools and Technologies

The implementation utilized the following technologies:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- NLTK
- Scikit-learn
- Joblib
- Regular Expressions (re)

These tools facilitate efficient data processing, feature engineering, model training, and deployment.

2.5 Data Preprocessing

To improve classification performance, several preprocessing techniques were applied:

1. Conversion to lowercase
2. Removal of punctuation
3. Tokenization
4. Stop-word removal
5. Stemming using Porter Stemmer
6. Noise reduction

These operations help standardize the text and reduce dimensionality.

2.6 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert textual data into numerical feature vectors.

Advantages of TF-IDF

- Highlights important terms

- Reduces impact of common words
- Produces sparse feature representations
- Enhances classification performance

2.7 Classification Model

Logistic Regression was selected due to:

- High efficiency in binary classification
- Fast prediction speed
- Low computational complexity
- Strong interpretability
- Excellent text classification performance

2.8 System Workflow

News Dataset

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Text Cleaning

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Tokenization

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Stop-word Removal

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Stemming

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TF-IDF Vectorization

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Train-Test Split (80:20)

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Logistic Regression

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Prediction

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Fake / Real Classification

3. Theoretical Background

3.1 Fake News Detection

Fake news detection is the process of automatically identifying misleading or fabricated information using computational methods. Modern systems leverage machine learning algorithms to recognize patterns associated with misinformation.

3.2 Natural Language Processing

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand and process human language. NLP techniques play a vital role in text classification and fake news detection.

3.3 TF-IDF

TF-IDF measures the importance of words within a document relative to a collection of documents. Terms that frequently appear in a document but rarely occur in the overall dataset receive higher weights.

3.4 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification. It estimates probabilities and predicts whether an input belongs to a particular class.

4. Experimental Results and Discussion

4.1 Dataset Split

The dataset was divided into:

- Training Data = 80%
- Testing Data = 20%

Training Samples ≈ 8,180

Testing Samples = 2,026

4.2 Classification Performance

Metric	Real News	Fake News
Precision	0.95	0.97

Recall	0.97	0.94
F1-Score	0.96	0.96

Overall Performance

- Accuracy = 95.95%
- Macro F1-Score = 96%
- Weighted F1-Score = 96%

The results indicate that the model effectively distinguishes between fake and real news articles with high accuracy.

4.3 Confusion Matrix

Actual / Predicted	Real	Fake
Real	1027	27
Fake	55	917

Interpretation

- Correctly Classified Real News = 1027
- Correctly Classified Fake News = 917
- False Positives = 27
- False Negatives = 55

The low number of misclassifications demonstrates the robustness of the model.

4.4 Discussion

The Logistic Regression classifier achieved a high accuracy of 95.95%, indicating strong capability in identifying deceptive news content. The balanced precision and recall values suggest that the model effectively minimizes both false positives and false negatives. The findings confirm that traditional machine learning methods remain highly competitive when combined with effective NLP preprocessing and feature engineering techniques.

5. Research Gap and Contributions

5.1 Research Gap

Most existing fake news detection systems rely on deep learning architectures such as CNNs, LSTMs, and Transformers. Although these models often achieve high performance, they require substantial computational resources and large training datasets.

Limited research has explored whether lightweight machine learning approaches can achieve comparable performance on multilingual translated news datasets.

5.2 Contributions

This project contributes by:

- Developing an automated fake news detection framework
- Utilizing a multilingual translated dataset
- Applying NLP preprocessing techniques
- Implementing TF-IDF feature engineering
- Achieving 95.95% accuracy
- Creating a deployable machine learning pipeline using Joblib

6. Applications

The proposed system can be applied in:

- Social Media Monitoring
- News Verification Platforms
- Online Journalism
- Government Information Systems
- Election Misinformation Detection
- Fact-Checking Organizations
- Educational Content Verification

7. Conclusion and Future Work

This research successfully developed a machine learning-based fake news detection system using Data Mining and Natural Language Processing techniques. The combination of TF-IDF feature extraction and Logistic Regression classification achieved an overall accuracy of 95.95%, demonstrating excellent effectiveness in identifying fake and real news articles.

The results prove that lightweight machine learning models can provide highly accurate fake news detection while maintaining low computational requirements. This makes the proposed framework suitable for deployment in real-world environments where efficiency and scalability are important.

Future work may focus on integrating Transformer-based models such as BERT, supporting multilingual native-language datasets, implementing Explainable AI (XAI) techniques, and deploying the system as a real-time web-based fact-checking platform.